



5 | AI-powered Driving Analytics: Understanding Driver Behavior through Data

Yash Tanaji Shinde^{1*}, Prathmesh Redekar² and Prakash K. Sonwalkar³

Driver behavior analysis (DBA) is one of the most important research topics in the field of driving analytics. In this chapter, we review the current state of the art of artificial intelligence (AI)–driven driving analytics, which is a disruptive way to perceive driver behavior as the application of powerful technologies to analyze and process the tremendous amounts of information gathered by vehicle sensors, telematics, and connected devices. The AI-powered driving analytics is based on deep learning and machine learning frameworks, such as deep learning, neural networks, and deep learning algorithms, which are able to identify and define driver behaviors. The data collected from in-vehicle sensors, vehicle trajectories, and mobile data helps to create reliable driver behavior detection systems. In addition, machine learning–based analytics are implemented to help control fuel consumption, improve road safety, and offer immediate feedback to the driver, which results in a quantifiable increase in driving efficiency and reduction of accidents. Adaptive traffic management systems are AI-operated solutions to organize traffic flow and increase the efficiency of public transport, being dynamic in adjusting to the challenges of the cities.

1. Introduction

Artificial intelligence (AI)–powered driving analytics is an advanced way to understand driver behavior through the application of powerful technologies to analyze and process the tremendous amounts of information gathered by the vehicle sensors, telematics, and connected devices. These advanced systems utilize machine learning and AI solutions in tracking the real-time driving patterns, quantifying and assessing the driver safety and automobile effectiveness to ensure minimum costs of operation without compromising the security of employees and other road users. AI systems will be able to draft a detailed description of each driver as they continually monitor aspects of vehicle speed, acceleration, braking dynamics, and steering habits to compile a detailed profile of any particular driver

¹⁻³ Department of Computer Science and Engineering, Jain College of Engineering and Research, Affiliated to Visvesvaraya Technological University, Belagavi, Karnataka, India.

* Corresponding author: syash0080@gmail.com

and find areas that need improvement. The technology has allowed motorists instant feedback concerning their performance to correct risky behaviors before they result in accidents and has promoted the culture of safe and efficient driving.

1.1 Evolution of Vehicle Telematics and Connected Car Technologies

Digitization, wireless networks, and cloud services have been the key factors that have led to the evolution of vehicle telematics technologies and the connected car realm [1]. Vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication is also implemented in connected vehicles, through which traffic safety and crash prevention are facilitated [2]. They combine cyber-physical systems (CPS) [3, 4] and the Internet of Things (IoT) to facilitate the transfer and exchange of information without any disturbances.[5, 6, 7]. Within the confines of automotive telematics, there are multiple applications, such as GPS, infotainment, and onboard diagnostics, which are among the most effective to use for enhancing navigation and vehicle performance [8, 9]. New models and opportunities for business and services have emerged due to the development of connected cars, including car-sharing and smart ticketing [1]. With the advancement of this technology, self-driving cars are becoming a reality, which implies that they will affect many industries, such as the insurance field and transportation services [5, 8].

1.2 The Emergence of Big Data in Transportation Systems

The concept of big data analytics is transforming intelligent transportation systems (ITS) and connected/automated vehicles (CAV), providing innovative insights into traffic safety, driver behavior, and system maintenance [10]. Now, with the availability of information derived in different ways (structured and unstructured data produced by sensors, text analytics, and image processing), more holistic analysis and prediction of traffic patterns, accidents, and driver behavior can be undertaken [3, 11, 12]. Big data analytics also plays a critical role in the maintenance of the transportation infrastructure in terms of using the Internet of Things (IoT) and sensors to gather and analyze large volumes of data and make efficient decisions about its limitations and constraints [14].

1.3 Growing Importance of DBA for Safety and Efficiency

The increasing significance of driver behavior analysis (DBA) for both safety and efficiency, according to an ITS, is realized in most of the contemporary studies. DBA utilizes the information in in-vehicle networks, sensors, and communication technology to deliver driving patterns [15]. This analysis is applied to different things, such as traffic safety, the development of autonomous vehicles, energy management, and driver profiling [16]. Innovative platforms such as ADAPT DBA

allow **them** to process the data on board efficiently and add social incentives to enhance the safety of the drivers [17, 18].

1.4 Limitations

The limitations of subjective driver evaluation techniques are variability in human judgments [19, 20], lack of reproducibility [21], integration challenges with objective metrics, and need for balanced approach.

1.5 Need for Objective, Data-driven Insights into Driving Patterns

Hard data insights into driving trends are being considered more and more necessary in terms of road safety, improved fuel efficiency, and more civil driver behavior. Analytics with the help of AI offer a sound system for realizing these patterns by analyzing huge volumes of driving data. Machine learning AI technologies allow us to dynamically analyze the driving behavior, and as a result, make improvements that lead to fuel consumption and minimal environmental impact [22].

Data-driven frameworks and predictive models: In predicting driving behaviors, a data-driven framework based on unsupervised learning has been reported to be highly accurate as compared to the traditional approaches. In-vehicle sensor data could be used to split driving styles effectively and gather a pattern that can guide safety operations, particularly with elderly drivers [23, 24, 25].

1.6 Privacy and Ethical Considerations in Driver Monitoring

The ethical and privacy issues associated with AI-powered driving analytics and self-driving cars are known to be severe. The need to balance between safety and privacy is acute because the amount of data collected and relayed by the AI system about the driver is enormous [26, 27]. Through the adoption of responsible research and innovation frameworks, such problems could be mitigated by engaging the stakeholders and making **it** socially acceptable [28, 29]. Algorithms of fairness and AI ethical use during development are vital in ensuring better decision-making and reduced risks to privacy.

1.7 Scope and Significance

Driving analytics applications use AI to analyze information from vehicle sensors, telematics, and highly sophisticated algorithms to evaluate and appreciate driver behavior. Some of the important uses of these analytics are:

- Insurance [30, 31, 32]
- Fleet management [33]
- Autonomous vehicle development [34–37]

In short, AI-based driving analytics are disrupting the automotive ecosystem with respect to safety, efficiency, and economic partnerships among insurers, fleet managers, technology providers, and their associates.

2. Methods

The specified data collection framework is aimed at recording a rich range of different driving behaviors, both with the help of innovative sensors and cheap data capture techniques, so that wide coverage is provided. Along with a solid data collection framework to ensure the appropriate analysis of driver behavior, AI-powered driving analytics is necessary. This framework incorporates numerous sensor technologies and sources of data that allow having detailed knowledge about driving patterns and behaviors [38], as shown in Figure 5.1. This design will take advantage of the smartphone sensors and other low-priced approaches to collect real-time data, improving the perception of driver behavior in naturalistic conditions [39, 40].

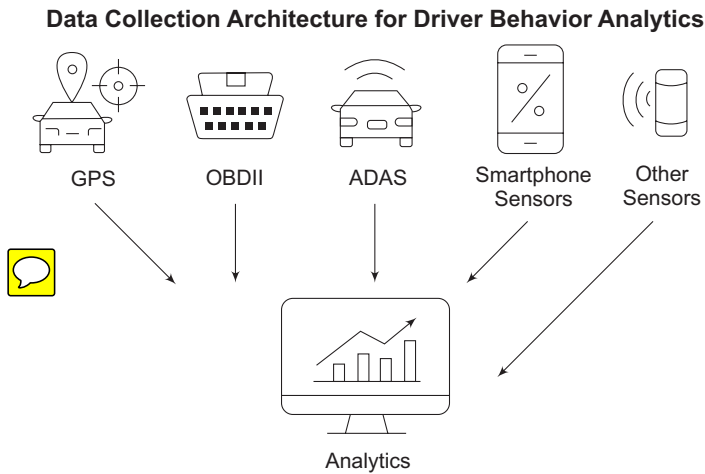


Fig. 5.1 Multimodal data collection architecture for driver behavior monitoring.

The framework uses sensor technologies and data sources which are GPS Tracking Systems and Location Analytic [41], Measurements of Accelerometer and Gyroscope [42, 43], Data Guided Sensing Applications [44–46], Systems Integration, Advanced Driver Assistance [47], and Computer Vision (CV) [34, 49]. A data type is the mode of information delivery. It defines the extent to which information delivery can be conveyed. Data types are either binary, integer, or real. Binary data type is the codification of information. The combination of such a variety of data sources results in the diversity of data types and parameters necessary to conduct a complete driving habit analysis, including kinematic data, spatial data [50, 51], temporal patterns [52], and environmental context [53, 54, 55].

The starting point for effectively managing data quality is developing a sound framework that will help ensure that the data collected from various sources is accurate, complete, and timely. This is achieved by adopting strict data capture

procedures and constant surveillance to detect any possible lapses in the quality process early enough [5].

- *Methods of Noise Filtration and Signal Filtering:* The smoothing of anomalies in the data presented by environmental conditions or inaccuracy in the operation of sensors is achieved by techniques like low-pass filters [57].
- *Imputation Strategies for Missing Data:* Missing data can highly distort the outcome of the analysis. To handle this, various methods can be used such as mean/mode imputation in a variable of numeric data, k-nearest neighbors, and regression-imputation strategies to impute missing information by using patterns [58].
- *The Procedures of Outlier Detection and Handling:* Outliers may be a representation of faulty data or rather one-of-a-kind habits of driving. The Z-score method and the interquartile range are some of the statistical tools that we apply to identify outliers [40, 60].

The key task of feature engineering is to transform raw data into usable variables that would improve the models. This involves behavioral pattern extraction algorithms [61], statistical feature computation [62], time-series analysis and segmentation [63], and spatial analytics and route optimization metrics [64].

2.1 AI and Machine Learning Approaches

It is critical to apply AI and machine learning approaches to develop meaningful interpretations of processed data and use them to support the right actions, as shown in Figure 5.2.

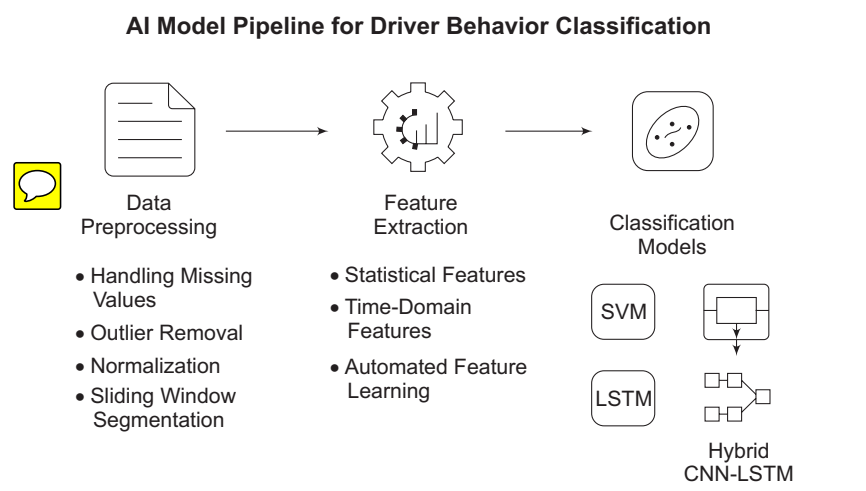


Fig. 5.2 Pipeline of AI-driven behavior analysis

2.2 Supervised Learning

The driver profiling models and risk prediction models are mainly based on supervised learning. The important methods include Classification Models to Profile Drivers [65], Regression on Risk Prediction [66], Support Vector Machine [67], and Spatial Analytics and Route Optimization Metrics [68].

2.3 Unsupervised Learning

The unsupervised learning methods are used to discover obscure data patterns without predetermined labels: Driver Segmentation with Clustering Algorithms [69] and Principal Component Analysis Dimensionality Reduction.

2.4 Deep Learning and High-Quality AI Techniques

Deep learning is used because of the potential to design very complex relationships within big datasets: Recurrent Neural Networks with Sequential Patterns, Long Short-Term Memory (LSTM) networks, transformer models, reinforcement learning, transfer learning, and explainable AI.

3. Analytics

3.1 Development of a Taxonomy of Behavior

A profound achievement of developing a detailed taxonomy of behaviors has been categorizing driver behavior into specific categories and achieving specific interventions. It has helped to detect cases of aggressive driving—such as quick acceleration, illegal braking, and frequent lane changes—through the use of strong machine learning algorithms.

3.2 Validating Performance and Verifications

The validations and the performance measures of these systems of classification can offer hopeful findings. The accuracies of classification vary across different drivers, but the precision and recall percentage rates of detecting behaviors are high in most cases.

3.3 Risk Assessment Models

The implementation of risk assessment models has yielded significant insights into accident probability prediction. By analyzing historical driving data and behavioral patterns, we have developed algorithms that predict the likelihood of accidents with a high degree of accuracy.

3.4 Behavioral Intervention Systems

Behavioral intervention systems have been designed to foster safer driving habits through real-time feedback mechanisms. These systems employ gamification strategies to encourage behavior modification, offering rewards for safe driving and personalized coaching recommendations tailored to individual drivers.

4. Discussion

The use of AI-powered driving analytics enables applying the data obtained from various sources to gain insight into the driving behavior that is concerned with the recognition of behavioral patterns. It implies determining important predictors of the driving behavior, the correlation analysis of different metrics, and the effect of the external factors. It also makes a difference between individual and population behavior analysis. The multimodal data analysis makes it possible to combine data collected from various sources and gain an overview of the driving behavior to identify complex relationships among metrics. Drive Sense enables the prediction of abnormal driving behaviors because it correlates findings with various driving patterns, thereby increasing the predictive capabilities.

5. Conclusion

The development of smart cities relies heavily on AI, with the connected infrastructure helping to provide data-based transportation services. Adaptive traffic management systems are AI-operated solutions to organize traffic flow, integrate smart city infrastructure, and increase the efficiency of public transport, being dynamic in adjusting to the challenges of the cities. The possibility of individual transportation optimization helps self-driving vehicles, focusing on the development of a collaborative ecosystem that fulfills individual mobility tasks. On the other hand, the fast development of AI in driving analytics can result in overdependence, reducing human driving capabilities and adversely affecting an individual's ability to be an adequate driver, thereby increasing their vulnerability in case of a system breakdown. It will be vital to strike a balance between innovation and caution to foster sustainability in this field. The operating field of AI-based driving analytics promises to achieve tremendous breakthroughs in the future, especially through fully automated driving systems, smart cities, customized commuting, and social implications.

References

- [1] Arena, F., et al. (2021). Future connected cars through the evolution of telematics and infotainment. *International Journal of Engineering*, 9, 49–61.
- [2] Narla, S.R. (2013). The evolution of connected vehicle technology: From smart drivers to smart cars to self-driving cars. *ITE Journal: Institute of Transportation Engineers*, 83, 2013.
- [3] Estrela, V.V., de Jesus, M.A., Intorne, A.C., Batista, K.K.S., Deshpande, A., Shi, F., Laghari, A.A., Khan, A.A. and Oliveira, L.P. (2023). Medical visual thernagnostic systems using artificial

- intelligence (AI): Principles and perspectives. In *Intelligent Healthcare Systems*, ed. V. V. Estrela, 301–321. CRC Press.
- [4] Estrela, V.V. and Deshpande (2023). Scale and resolution issues regarding medical images: Challenges ahead. In *Intelligent Healthcare Systems*, ed. V. V. Estrela, 90–109. CRC Press.
- [5] Möller, D.P.F. and Haas, (2019). The connected car. In *Guide to Automotive Connectivity and Cybersecurity*.
- [6] Jumani, A.K. and Laghari, (2025). Image processing-based via agridrones for agriculture fields. In *Intelligent Designs, Innovations and Sustainability in Agriculture 4.0*, eds A. Deshpande, V.V. Estrela and A. Khan. CRC Press
- [7] Estrela, V.V., Jumani, A.K. and Laghari, (2023). Internet of Medical Things layers for medical cyber-physical systems. In *Intelligent Healthcare Systems*, ed. V.V. Estrela, 45–65. CRC Press.
- [8] Belafonte-Young, D. (2018). Automotive telematics. In *Communication Technology Update and Fundamentals*.
- [9] Estrela, V.V. and Deshpande, (2023). Ad hoc networks in healthcare intelligent transportation systems humanitarian relief: MANETs, VANETs, and FANETs. In *Intelligent Healthcare Systems*, ed. V.V. Estrela, 66–89. CRC Press.
- [10] Zhu, L., et al. (2019). Big data analytics in intelligent transportation systems: A survey. *IEEE Transactions on Intelligent Transportation Systems*, 20, 383–398.
- [11] Kulkarni, D.S. and Rodd, S.F. (2018). Extensive study of text-based methods for opinion mining. *IEEE International Conference on Inventive Systems and Control (ICISC)*, 523–527
- [12] Fathi, F., Abghour, N. and Ouzzif, M. (2018). From big data to better behavior in self-driving cars. In *Proceedings of the 2nd International Conference on Cloud and Big Data Computing*, 42–46.
- [13] Kour, R., et al. (2019). Big data analytics for maintaining transportation systems. *Asset Analytics*.
- [14] Nasr Azadani, M. and Boukerche, A.F.M. (2021). Driving behavior analysis guidelines for intelligent transportation systems. *IEEE Transactions on Intelligent Transportation Systems*, 23, 6027–6045.
- [15] Liu, J., et al. (2021). A survey of recent advances in driving behavior analysis. In *Proceedings of the 3rd International Symposium on Smart and Healthy Cities*, 145–157.
- [16] Kargupta, H. and Kargupta, P. (2019). Analyzing driving data using the ADAPT distributed analytics platform for connected vehicles. *IEEE International Conference on Data Science and Advanced Analytics*, 590–599.
- [17] Bouhoute, A. (2019). Advanced driving behavior analytics for an improved safety assessment and driver fingerprinting. *IEEE Transactions on Intelligent Transportation Systems*, 20, 2171–2184.
- [18] Nybacka, M., et al. (2014). Links between subjective assessments and objective metrics for steering, and evaluation of driver ratings. *Vehicle System Dynamics*, 52(Suppl. 1), 31–50.
- [19] Voß, G.M.I., Keck, C.M. and Schwalm, M. (2018). Investigation of drivers' thresholds of a subjectively accepted driving performance with a focus on automated driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 56, 280–292.
- [20] Chai, W., et al. (2024). Rethinking the evaluation of driver behavior analysis approaches. *IEEE Transactions on Intelligent Transportation Systems*, 25(8), 9958–9966.
- [21] Haghsheenas, S.S., et al. (2024). Artificial intelligence-powered driver behavior analysis for fuel consumption optimization: A pathway to greener roads. In *Proceedings of the 10th International Conference on Control, Decision and Information Technologies*, 116–122.
- [22] Milardo, S., et al. (2022). A data-driven framework for driving style classification. In *Proceedings of the International Conference on Advanced Data Mining and Applications*.
- [23] Moshfeghi, S. and Jang, J. (2024). Pattern mining of older drivers' driving behavior through telematics-data-driven unsupervised learning. *Authorea Preprints*.

- [24] Abdelrahman, A.E., Hassanein, H.S. and Abuali, N.A. (2022). Robust data-driven framework for driver behavior profiling using supervised machine learning. *IEEE Transactions on Intelligent Transportation Systems*, 23, 3336–3350.
- [25] Krontiris, I., et al. (2020). Autonomous vehicles: Data protection and ethical considerations. In *Proceedings of the 4th ACM Computer Science in Cars Symposium*.
- [26] Oueida, S., et al. (2021). A fair and ethical healthcare artificial intelligence system for monitoring driver behavior and preventing road accidents. *arXiv*, 2107.
- [27] Stahl, B.C. and Wright, D. (2018). Ethics and privacy in AI and big data: Implementing responsible research and innovation. *IEEE Security & Privacy*, 16, 26–33.
- [28] Kumar, A. (2024). Exploring ethical considerations in AI-driven autonomous vehicles: Balancing safety and privacy. *Journal of Artificial Intelligence General Science*.
- [29] Bhattacharya, S., et al. (2025). AI revolution in insurance: Bridging research and reality. *Frontiers in Artificial Intelligence*, 8.
- [30] Sun, S., et al. (2020). Assessing driving risk using Internet of Vehicles data: An analysis based on generalized linear models. *Sensors*, 20.
- [31] Hanafy, M. and Ming, R. (2021). Machine learning approaches for auto insurance big data. *Risks*.
- [32] Edymond, H.R. (2024). Advanced data science applications in vehicles: A comprehensive review. *International Journal of Technology and Systems*.
- [33] Driessen, T., et al. (2024). How AI from automated driving systems can contribute to the assessment of human driving behavior. *Robotics*.
- [34] Atakishiyev, S., et al. (2021). Explainable artificial intelligence for autonomous driving: A comprehensive overview and field guide for future research directions. *IEEE Access*, 12, 101603–101625.
- [35] Yun, K., et al. (2024). A study on machine learning-enhanced roadside unit-based detection of abnormal driving in autonomous vehicles. *Electronics*.
- [36] Razmjoo, N., Deshpande, A., Khalilpour, M., Estrela, V.V., Padilha, R. and Monteiro, A.C.B. (2021). Optimal bidding strategy for power market based on improved world cup optimization algorithm. In *Metaheuristics and Optimization in Computer and Electrical Engineering*, Vol. 696, eds N. Razmjoo, M. Ashourian, & Z. Foroozandeh.
- [37] Tong, J. and Ren, T. (2024). Research on multimodal intelligent navigation and AI+AR display design theory. *Journal of Research in Science and Engineering*, 6(8).
- [38] de Almeida, M.B. (2023). AI-based framework for automatically extracting high–low features from NDS data to understand driver behavior.
- [39] Farfan, W.S. and Saotome, (2020). Integrated optical flow for situation awareness, detection and avoidance systems in UAV systems. In *Imaging and Sensing for Unmanned Aircraft Systems*, Vol. 1, eds V.V. Estrela, J. Hemanth, O. Saotome, G. Nikolakopoulos and R. Sabatini, 47–74. IET.
- [40] Mehta, A., et al. (2023). Securing the future: A comprehensive review of security challenges and solutions in advanced driver assistance systems. *IEEE Access*, 12, 643–678.
- [41] Kapilan, K., Bandara, S. and Dammalage, T. (2020). Vehicle accident detection and warning system for Sri Lanka using GNSS technology. In *Proceedings of the International Conference on Image Processing and Robotics*, 1–5.
- [42] Yaqoob, S., et al. (2024). AI-driven driver behavior assessment through vehicle and health monitoring for safe driving—A survey. *IEEE Access*, 12, 48044–48056.
- [43] Laghari, A. A., et al. (2024). Quality of experience assessment in virtual/augmented reality serious games for healthcare: A systematic literature review. *Technology and Disability*, 36, 17–28.
- [44] Jumani, A.K., Shi, J., Laghari, A.A., Estrela, V.V., Sampedro, G.A., Almadhor, A., Kryvinska, N. and ul Nabi, A. (2024). Quality of experience that matters in gaming graphics: How to blend image processing and virtual reality. *Electronics*, 13(15), 2998.

- [45] Kondapaka, K.K. (n.d.). Integrating AI and augmented reality for enhanced driver assistance and navigation in modern vehicles. *Journal of Machine Learning for Healthcare Decision Support*.
- [46] Guirrou, H., Es-sadek, M. and Taher, Y. (2023). Artificial intelligence in self-driving: Study of advanced current applications. *International Journal of Engineering Trends and Technology*, 71(8), 225–242.
- [47] Deshpande, A., Estrela, (2023). Super-resolution in a world of scarce resources for medical imaging applications. In *Edge-AI in Healthcare: Trends and Future Perspectives*, eds S. Vyas et al.. CRC Press.
- [48] Musicant, O., et al. (2017). Integrating kinematic- and vision-based information to better understand driving behaviour. In *Proceedings of the International Driving Symposium on Human Factors in Driver Assessment, Training and Vehicle Design*, 6, 518–524.
- [49] Chanmas, G., et al. (2024). Driving assistant using generative AI pre-generated messages in simulator-based driving assessment: A step towards low-cost simulator-based driving assessment. *Heliyon*, 10(16).
- [50] Sharma, A., et al. (2023). *Autonomous Vehicles: Challenges and Advancements in AI-based Navigation Systems*.
- [51] Capitaine, M. and Cárdenas, H. (2023). Artificial intelligence and advanced driver assistance systems absorption in Mexico. *Ciencia Nicolaita*, 88.
- [52] Baccari, S., et al. (2024). Anomaly detection in connected and autonomous vehicles: A survey, analysis, and research challenges. *IEEE Access*, 12, 19250–19276.
- [53] Langner, L., et al. (2019). A driving scenario representation for scalable real-data analytics with neural networks. In *Proceedings of the IEEE Intelligent Vehicles Symposium*, 2215–2222.
- [54] Shih, Y.-T. and Chen, P.-C. (2013). *System and Method of Reducing Noise* (Patent).
- [55] Amir, M. and Jensen, R. (2016). Missing data imputation using fuzzy-rough methods. *Neurocomputing*, 205, 152–164.
- [56] Hoque, A. and Ahmed, (2025). AI-assisted imaging and navigation in minimally invasive cardiac interventions: A systematic review. *Insights—Journal of Health and Rehabilitation*.
- [57] Chougule, A. and Chamola, (2023). A comprehensive review on limitations of autonomous driving and its impact on accidents and collisions. *IEEE Open Journal of Vehicular Technology*, 1–20.
- [58] Vedantham, A.K. (2024). Augmented reality in modern vehicles: Enhancing safety and driver experience. *International Journal for Multidisciplinary Research*, 6(6).
- [59] Yamaguchi, Y. (2024). *AR-Navi: An in-vehicle navigation system using video-based augmented reality technology*, 2024.
- [60] Cao, J., et al. (2010). Analysis and application of the optimal path based on ArcGIS geodatabase network model. In *Environmental Science and Information Application Technology*, Vol. 1, 434–437.
- [61] Halim, Z., Kalsoom, R. and Baig, A.R. (2016). Profiling drivers based on driver-dependent vehicle driving features. *Applied Intelligence*, 44(3), 645–664.
- [62] Hauer, E. (2015). *The art of regression modeling in road safety*.
- [63] Yeo, M.V.M., et al. (2009). Can SVM be used for automatic EEG detection of drowsiness during car driving? *Safety Science*, 47(1), 115–124.
- [64] Oguchi, K. and Weir, D.F.R. (2008). *System for predicting driver behavior* (US Patent No. US20080167820A1).
- [65] Fu, Z., Hu, W. and Tan, T. (2005). Similarity-based vehicle trajectory clustering and anomaly detection. In *Proceedings of the International Conference on Image Processing*, Vol. 2, 602–605.